Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2023
Course Code: EEE 421, Session: 2019-2020
Course Title: Solid State Devices

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) Describe three classifications of solids-amorphous, polycrystalline, and single crystal. 6
   (b) The lattice constant of a face-centered cubic lattice is 0.425 nm. Determine the 6
      i. effective number of atoms per unit cell
      ii. volume density of atoms
   (c) If the free electrons in a semiconductor have effective mass $m_c = 0.1m_o$, where m_o is the mass of the electron in vacuum, what is their average speed at $T = 300K$? If the free holes in the same material have effective mass $m_v = 1.5m_o$, what is their average speed at the same temperature? Assume a Maxwell-Boltzmann distribution of the electron energies. 7
   (d) Explain Schottky barrier. How it differs from ohmic contact? Explain. 6

2. (a) Starting from charge neutrality condition, Find thermal equilibrium carrier concentration at compensated semiconductor in terms of N_a, N_d and n_i 6
   (b) Consider silicon at $T = 300$ K. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm and $L = 0.50$ cm. The electrical parameters measured are: $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ *gauss* $= 6.5 \times 10^{-2}$ *tesla*. The Hall field is $E_H = -16.5$ mV cm^{-1}. Determine: 8
      i. the Hall voltage
      ii. the conductivity type
      iii. the majority carrier concentration and
      iv. the majority carrier mobility
   (c) Using proper graph, explain how carrier concentration varies with temperature. 5
   (d) Sketch and briefly describe energy band diagram of compensated semiconductor. 6

3. (a) Sketch and explain density of states functions, Fermi distribution, and carrier concentration distribution curves of n-type material in a single graph. 8
   (b) Explain how barrier potential maintains equilibrium p, n regions. 4
   (c) Explain dielectric relaxation time. Find the equation for dielectric relaxation. 7
   (d) Consider a compensated n-type silicon at $T = 300$ K with a conductivity of ρ [*sic*: should be σ] $= 16\ (\Omega - \text{cm})^{-1}$ and an acceptor doping concentration of 10^{17} cm^{-3}. Determine the doner [*sic*: should be donor] concentration and the electron mobility. 6

Part-B

*[Answer any **two** of the following questions]*

4. (a) With proper graphs, explain how mobility is affected by temperature and doping concentration. 6

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(b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. 7

(c) For a pn junction with $N_a = 1.5 \times 10^{18}\text{cm}^{-3}$, $N_d = 1.5 \times 10^{16}\text{cm}^{-3}$ and $n_i = 2 \times 10^{10}\text{cm}^{-3}$ 6

i. calculate the built-in potential barrier V_{bi}
ii. calculate the depletion region with [*sic*: should be width] and E_{max} (Assume $\epsilon_s = 11.7\epsilon_o$) and
iii. what will be the width at liquid nitrogen temperature (80K).

(d) With proper graphs, explain time and spatial dependence of excess carrier concentration. 6

5. (a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will being [*sic*: should be will begin] diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. 7

(b) Find ideal pn junction current and sketch ideal electron and hole current components. 8

(c) Consider an n-type semiconductor at $T = 300$ K with carrier concentrations of $n_o = 10^{15}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3} and $p_o = 10^5$ cm^{-3}. In non-equilibrium, assume that the excess carrier concentrations are $\delta n = \delta p = 10^{-13}$ [*sic*: should be 10^{13}] cm^{-3}. Calculate the quasi-Fermi energy levels. 4

(d) An abrupt silicon pn junction has dopant concentrations of $N_a = 2 \times 10^{16}$ cm^{-3} and $N_d = 2 \times 10^{15}$ cm^{-3} at $T = 300$ K. Calculate: 6

i. V_{bi}
ii. W at $V_R = 0$ and $V_R = 8$ V and
iii. the maximum electric field in the space charge region at $V_R = 0$ and $V_R = 8$ V

6. (a) The body of a nanoscale N-channel MOSFET is doped at $N_A = 5^{18}$ [*sic*: should be 5×10^{18}] cm^{-3}. 12

i. Determine the maximum inversion-layer charge per unit area (Q_I) that can be reached so that the electric field in the oxide remains below $E_{ox} = 0.6$ V/nm. Neglect any interface-trap and gate-oxide charge.
ii. If the channel length and the channel width are $L = 50$ nm and $W = 500$ nm, respectively, determine how many electrons are creating this inversion layer.
iii. Calculate the threshold voltage of a MOSFET having gate-oxide thickness $t_{ox} = 1$ nm and flat-band voltage $V_{FB} = 1.0$ V.
iv. What gate voltage is needed to form the maximum inversion-layer charge?

(b) Determine the mode of operation of an NPN BJT with $\beta \approx 450$, if it is known that: 6

i. $V_{BE} = 0.7V$, $V_{CE} = 5.2V$
ii. $V_{BE} = 0.7V$, $V_{CE} = 0.2V$
iii. $V_{BE} = 0.8V$, $V_{BC} = 0.3V$
iv. $V_{BE} = -0.8V$, $V_{BC} = 0.7V$
v. $I_C = 455mA$, $I_B = 1mA$
vi. $I_C = 455mA$, $I_E = 502mA$

(c) Mention some advantages of using MOSFET. 3

(d) Does the doping level in the base of a BJT affect the emitter efficiency γ_E? If so, why? 4

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Some useful equations

$f_F(E) = \dfrac{1}{1 + exp\left(\frac{E-E_F}{kT}\right)}$	$g_c(E) = \dfrac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$
$\epsilon_{Si} = 11.7\epsilon_0$	$\dfrac{m^*}{m_0} = 0.26(P\ at\ Si)$
$n_0 = N_c\ exp\left[\dfrac{-(E_c - E_F)}{kT}\right]$	$n_0 = n_i\ exp\left[\dfrac{E_F - E_{Fi})}{kT}\right]$
$n_i^2 = N_c N_v exp\left[-\dfrac{E_c - E_v}{kT}\right] = N_c N_v exp\left[-\dfrac{E_g}{kT}\right]$	$n_d = \dfrac{N_d}{1 + \frac{1}{2}exp\left(\frac{E_d - E_F}{kT}\right)}$
$n_0 = \dfrac{N_d - N_a}{2} + \sqrt{\dfrac{(N_d - N_a)^2}{2} + n_i^2}$	$E_F - E_{Fi} = kT\ ln\left(\dfrac{n_0}{n_i}\right)$
$J = en\mu_n E + ep\mu_p E + eD_n\Delta n - eD_p\Delta p$	$W = \left\{\dfrac{2\epsilon_s V_{bi}}{e}\left[\dfrac{N_a + N_d}{N_a N_d}\right]\right\}^{\frac{1}{2}}$
$D_p\dfrac{\partial^2(\delta p)}{\partial x^2} - \mu_p\left(E\dfrac{\partial(\delta p)}{\partial x} + p\dfrac{\partial E}{\partial x}\right) + g_p - \dfrac{p}{\tau_{pt}} = \dfrac{\partial(\delta p)}{\partial t}$ $D_n\dfrac{\partial^2(\delta n)}{\partial x^2} + \mu_n E\dfrac{\partial(\delta n)}{\partial x} + g' - \dfrac{\delta}{\tau_{n0}} = \dfrac{\partial(\delta n)}{\partial t}$	$\dfrac{\partial p}{\partial t} = -\dfrac{\partial F_p^+}{\partial x} + g_p - \dfrac{p}{\tau_{pt}}$ $V_{bi} = \dfrac{kT}{e}ln\left(\dfrac{N_a N_d}{n_i^2}\right)$
$\delta p(x,t) = \dfrac{e^{-\frac{t}{\rho_{p0}}}}{(4\pi D_p t)^{1/2}}exp\left[\dfrac{-(x - \mu_p E_0 t)^2}{4D_p t}\right]$	$x_n = \left\{\dfrac{2\epsilon_s V_{bi}}{e}\left[\dfrac{N_a}{N_d}\right]\left[\dfrac{1}{N_a + N_d}\right]\right\}^{1/2}$
$E_{max} = -\dfrac{e}{\epsilon_s}\left\{\dfrac{2\epsilon_s V_{bi}}{e}\dfrac{N_a N_d}{N_a + N_d}\right\}^{\frac{1}{2}}$	$C' = \left\{\dfrac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right\}^{\frac{1}{2}}$
$\delta n_p(x) = n_p(x) - n_{p0} = n_{p0}\left[exp\left(\frac{eV_a}{kT}\right) - 1\right] exp\left(\frac{x_p + x}{L_n}\right)$	$p_n(x_n) = p_{n0}\ exp\left(\frac{eV_a}{kT}\right)$
$J = J_p(x_n) + J_n(-x_p) = \left[\dfrac{eD_p p_{n0}}{L_p} + \dfrac{eD_n n_{p0}}{L_n}\right]\left[exp\left(\frac{eV_a}{kT}\right) - 1\right]$	
$\delta n_B(x) \approx \dfrac{n_{B0}}{x_B}\left\{\left[exp\left(\dfrac{eV_{BE}}{kT}\right) - 1\right](x_B - x) - x\right\}$	$\delta p_C(x'') \equiv p_{C0}\ exp\left(-\dfrac{x''}{L_C}\right)$
$\delta p_E(x') \approx \dfrac{p_{E0}}{x_E}\left[exp\left(\dfrac{eV_{BE}}{kT}\right) - 1\right](x_E - x')$	

Term Test 1 (its questions (a) to (d) are 7(a) to 7(d))

[EEE 421 (2019-20) TT1, Time: 40 minutes, Marks: $5 \times 4 = 20$*]*

7. (a) Starting from charge neutrality condition, derive expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. 5

   (b) With proper equation and graph, explain variation of Fermi energy level in extrinsic semiconductor. 5

   (c) A special semiconductor material is to be "designed". The semiconductor is to be n-type and doped with $1 \times 10^{15} cm^{-3}$ donor atoms. Assume complete ionization and assume $N_a = 0$. The effective density of states functions are given by $N_c = N_v = 1.5 \times 10^{19} cm^{-3}$ and are independent of temperature. A particular semiconductor device fabricated with this material requires the electron concentration to be no greater than $1.01 \times 10^{15} cm^{-3}$ at $T = 400\ K$. What is the minimum value of the bandgap energy? 5

   (d) (i) Calculate the probability of an electron to be occupied at the lowest energy state of the conduction band of silicon when the Fermi energy is 0.3 eV below the conduction band at room temperature. (ii) Calculate the thermal electron concentration n_0. (iii) What

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is the thermal hole concentration p_0? (Room temperature N_c is 2.8×10^{19} cm^{-3}, n_i is 1.5×10^{10} cm^{-3}) 5

Term Test 2 (its questions (a) to (c) are 8(a) to 8(c))

[EEE 421 (2019-20) TT2, Time: 35 minutes, Marks: 6+7+7 = 20]

8. (a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. 6

   (b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. 7

   (c) Briefly describe different types of generation and recombination processes. Use appropriate diagram if necessary. 7

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